USC – AME532a – Assignment 6

Instructor: Dr. John McArthur

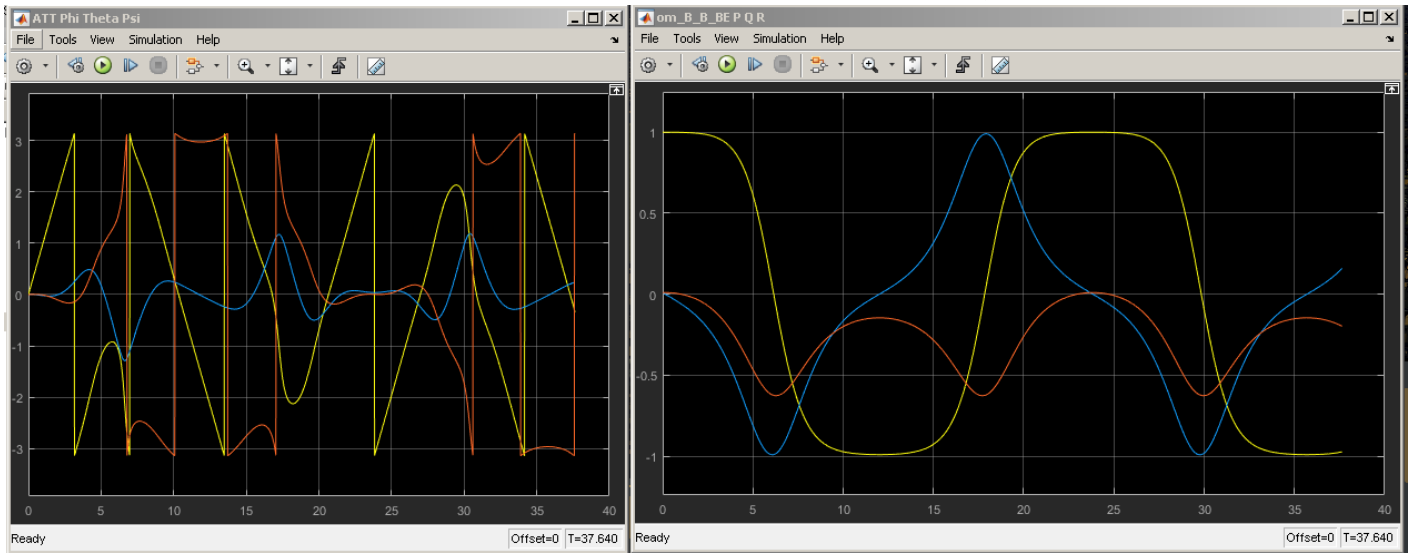
1) [50 points – 10 for each case]
For project, create a model where Forces and Moments from the aero-surfaces (in body axes) drive the equations of motion. Create another model where the Forces and Moments from the aero-surfaces are simply scoped/displayed/logged, and an Fext_B and Mext_B are used to drive the equations of motion. Call this a "tester" simulation. Run the following test cases and show results that validate that all parts are working. For your own tests, consider motor increasing RPM. Consider looking for roll damping. Consider giving beta and looking for yaw moment. Consider creating a turn with bank and up force.

When you are convinced that the model is working as expected, show results of a fully functional 6dof aircraft simulator where your Forces and Moments are driving you equations of motion. The aircraft (as defined above) should be stable for v_B_B_BE_0 = [13;0.25;0.25]; att_E_E_BN_0_deg<[10;10;any]; and om_B_B_BE_0<[1e-2;1;1e-2]. The aircraft will not return to zero roll, but will settle to a fairly small roll angle when starting from reasonable initial conditions.

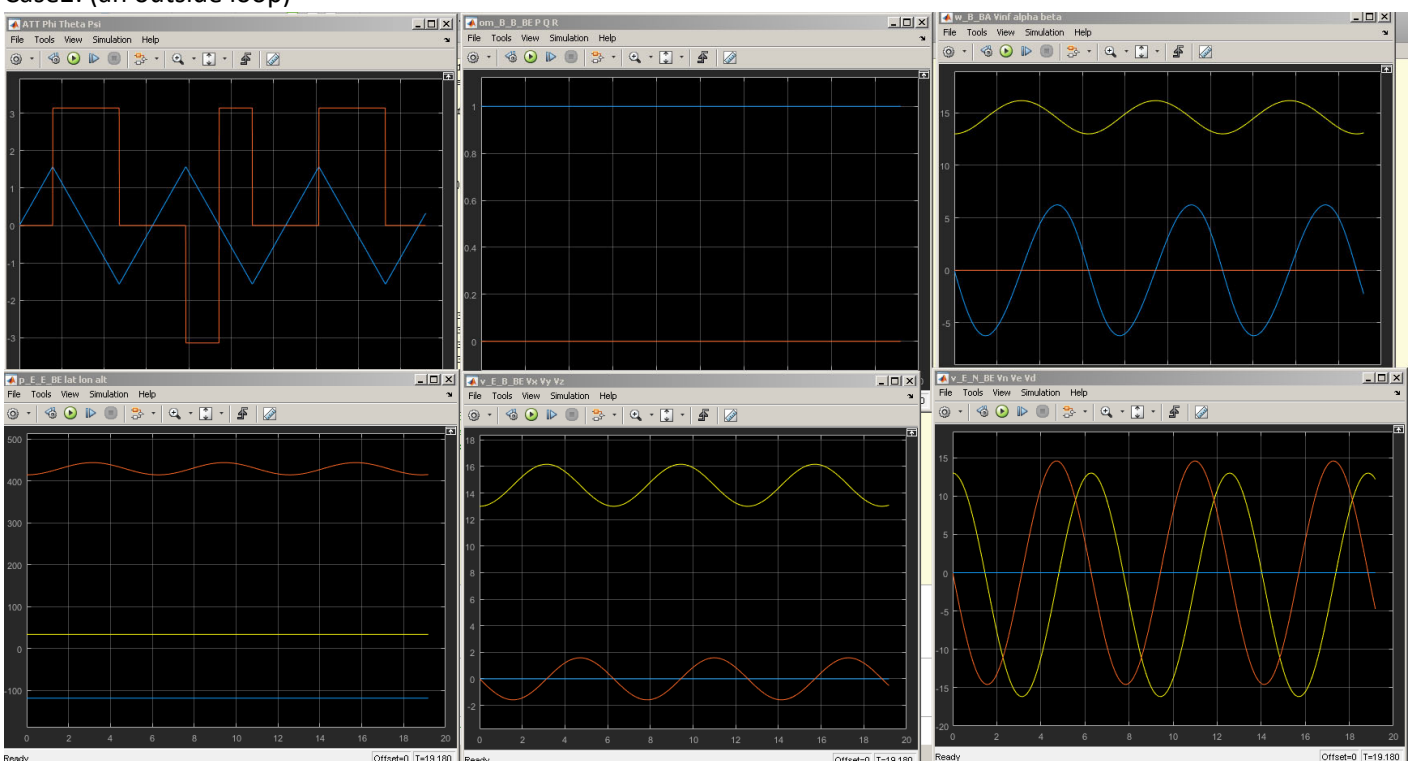
Test Case:	1	2	3	4	5
Name:	UnstableAxis	OutsideLoop	ForceField	YouDefine	YouDefine
p_E_E_BE_0:	[0;0;Rearth+h] or any	[0;0;Rearth+h] or any	[0;0;Rearth+h] or any		
v_B_B_BE_0:	[0;0;0]	[13;0;0]	[13;0;0.25]		
att_E_E_BN_0:	[0;0;0] or any	[0;0;0] or any	[0;0;0] or any		
om_B_B_BE_0:	[1;1e-2;1e-2]	[0;1;0]	[0;0;0]		
gravity:	0	0	-9.81		
Fext_B:	[0;0;0]	[0;0;-7]	[0;0;-4.7088]	FromAero	FromAero
Mext_B:	[0;0;0]	[0;0;0]	[0;0;0]	FromAero	FromAero
omMotor_0:	0	0	0		
omDotMotor:	0	0	0		

Some expected results to help you know what to expect: (yellow is first (x), blue is second (y), red is third (z))
I'd like to see your results with some interpretation and better formatting than mine.

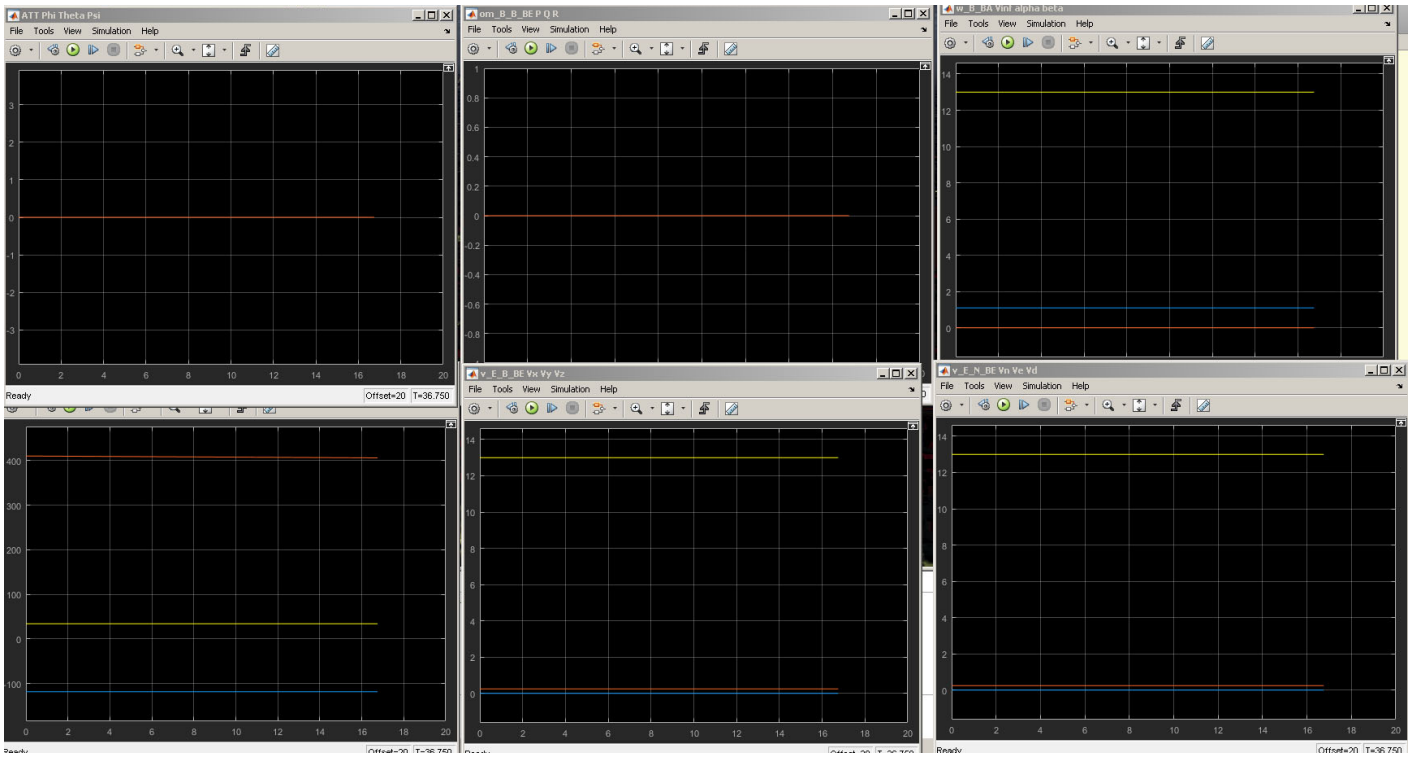
Case 1: (rotation about the unstable axis, with some rotation about the other axes)

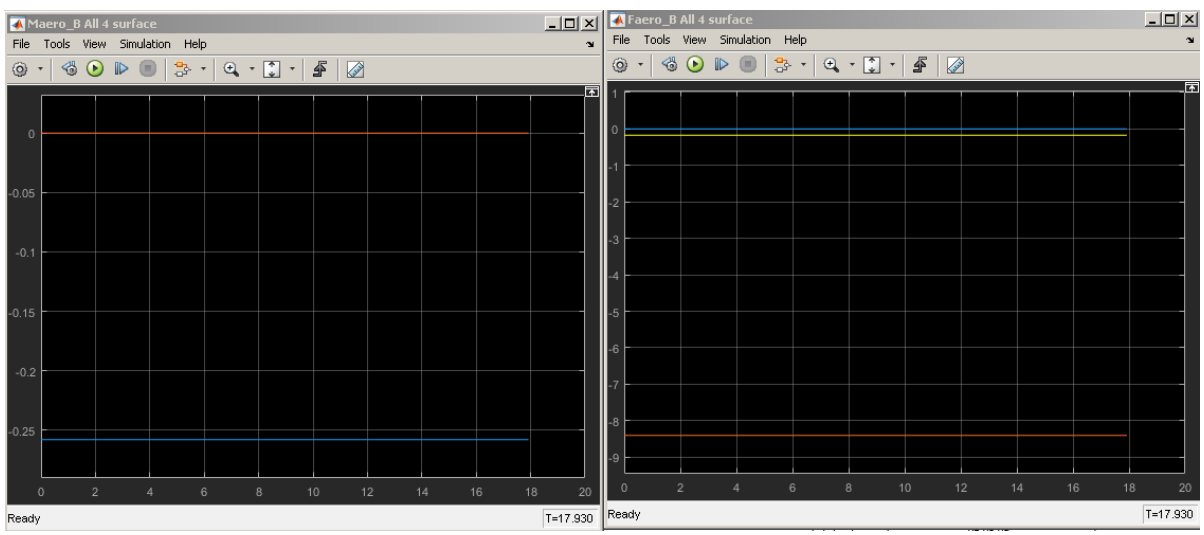


Case2: (an outside loop)



Case 3: ForceField test for aero surfaces check:





Ensure that your simulation is making sense and delivering reasonable results. If you need to make updates to it, show me what you are doing and explain them. If you think your simulation is running correctly and looking good, state so and provide some explanation/demonstration.

2) [10 points]
Continue to work on a plan for visualization of your results for your project. If you can pursue FlightGear, that is great. You can use Matlab's Instrumentation Control Toolbox to create a UDP object and send UDP data over to FlightGear. Or you can find my S-Function example on the class website to create UDP sockets in Simulink on your own. Also, scopes can be good, but difficult to see what is going on and understand if its working correctly or not. Putting data out "To Workspace" and plotting after the simulation is a good way to go. Can plot ground track and strip charts. Can

also create a 3D representation of your plane (stick figure plane is okay) and draw with roll, pitch, yaw for a good animation/visualization. Show some work/progress.